Time, Speed and Distance

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↳ The main important entity in time, speed & distance is speed. Based on kind of speed, this concept is categorized as.

→ Time & Distance.

→ Trains

→ Boats & Streams.

↳ The importance of time, speed & distance is to find one of the entity of the 3, when other two are given.

↳ Basic formula in this is

$$\text{Speed [kmph/mps]} = \frac{\text{Distance [km/m]}}{\text{Time [hr/sec]}}$$

↳ Conversion factor from kmph to mps:-

$$1 \text{ kmph} = \frac{1 \text{ km}}{1 \text{ hr}} = \frac{1000 \text{ m}}{60 \times 60 \text{ sec}} = \frac{5}{18} \text{ m/sec [mps]}$$

$$\therefore 1 \text{ kmph} = \left(\frac{5}{18}\right) \text{ mps}$$

↳ Conversion factor from mps to kmph:-

$$1 \text{ kmph} = \frac{5}{18} \text{ mps}$$

$$\Rightarrow 1 \text{ mps} = \left(\frac{18}{5}\right) \text{ kmph}$$

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↳ Proportionality Rules:-



(i) When Speed Constant:- $D \propto T \Rightarrow \frac{D_1}{D_2} = \frac{T_1}{T_2}$

(ii) When Distance Constant:- $S \propto \frac{1}{T} \Rightarrow \frac{S_1}{S_2} = \frac{T_2}{T_1}$

(iii) When Time Constant:- $S \propto D \Rightarrow \frac{S_1}{S_2} = \frac{T_2}{T_1}$

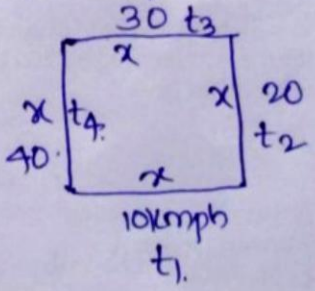
↳ Average Speed:- When total distance is covered in terms of individual distances with individual speeds then Average Speed is to be considered.

$$\text{Average Speed} = \frac{\text{Total distance Covered}}{\text{Total time}}$$

→ When an equal distance is covered twice with 2 individual speeds x & y then Average speed is $\left[\frac{2xy}{x+y}\right]$

$$A.S = \frac{d+d}{\frac{d}{x}+\frac{d}{y}} = \frac{2d}{\frac{d(x+y)}{xy}} = \frac{2xy}{x+y}$$

Ex 1:- A square shaped grass land is covered in terms of sides with speeds 10kmph, 20kmph, 30kmph & 40kmph. The Average Speed is?



$$A.S = \frac{\text{Total distance } (x+x+x+x)}{\text{Total time } (t_1+t_2+t_3+t_4)}$$

$$= \frac{4x}{\frac{x}{10}+\frac{x}{20}+\frac{x}{30}+\frac{x}{40}}$$

$$= \frac{4x}{\frac{12x+6x+4x+3x}{120}} = 4x \times \frac{120}{25x} \; \frac{4.8}{1} = \boxed{19.2 \text{ kmph}}$$

* Average of speeds here is $\frac{10+20+30+40}{4} = 25$ kmph.

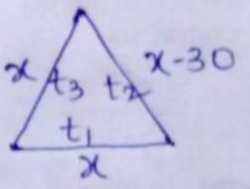
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* Average Speed is different from Average of Speeds except [when total distance is covered in terms of individual distances with equal speeds.] → **

Average Speed ≠ Average of Speeds [Except **]

Ex: 2:- Example for Avg Speed = Avg of Speeds:-

An equilateral star shaped land circulated in terms of sides with speeds 30kmph, 30 kmph & 30kmph. Then Avg speed & Avg of Speeds?



$$\text{Avg Speed} = \frac{\text{T.d}}{\text{T.t}} = \frac{x+x+x}{t_1+t_2+t_3}$$

$$= \frac{3x}{\frac{x}{30}+\frac{x}{30}+\frac{x}{30}} = \frac{3x}{\frac{3x}{30}} = 3\cancel{x} \times \frac{30}{3\cancel{x}} = \underline{30 \text{ kmph}}$$

$$\text{Avg of Speeds} = \frac{30+30+30}{3} = 30 \text{ kmph}.$$

here Avg of Speeds = Avg Speed

Ex: 3:- With $\frac{5}{6}$th of Speed A bus takes 10 mins more to reach the destination. The original time is?

$$\frac{5}{6}\begin{pmatrix} S \longrightarrow t \\ \frac{5}{6}S \longrightarrow t+10\text{mins} \end{pmatrix} \times\frac{6}{5} \Rightarrow \frac{6}{5}t = t+10\text{mins} \Rightarrow$$

$$\frac{6}{5}t - t = 10\text{mins} \Rightarrow \frac{1}{5}t = 10\text{ mins}$$

$$\boxed{t = 50 \text{ mins}}$$

Ex: 4:- Sandeep while going to airport with a speed of 60kmph. After completion of work on return journey he travelled with 20 kmph. The Avg Speed is?

$$x = 60, \; y = 20$$

$$\text{A.S} = \frac{2xxy}{x+y} = \frac{2\times60\times20}{60+20} = \frac{\overset{30}{\cancel{60}}\times40}{\underset{2}{\cancel{80}}} = 30 \text{ kmph}$$

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Ex:5:- A non-stop bus is moving with a speed of 120kmph. If stoppages are included its speed reduced to 80kmph Then stoppages time is?

Ans:- Reduction in speed due to stoppages = 120-80 = 40kmph

Thus 40kmph of speed spent on stoppages.

$\therefore$ Stoppages time = $\frac{\text{diff in speeds}}{\text{Original}}$ hr's.

$= \frac{120-80}{120} = \frac{40}{120} = \frac{1}{3}$ hr

Stoppages time = $\frac{1}{3}$ hr $= \frac{1}{3} \times 60 =$ 20 mins

↳ Relative Speed:- When 2 objects with speeds x & y are moving in opposite direction (or) same direction. to find out time taken to meet/time taken to cross one another, Relative Speed is used. Then The ~~relative speed~~ relative speed is to be considered.

[When the distance b/w 2 objects is 'd']

(i) → Relative Speed in opposite direction:-

Relative Speed in opposite direction = $x+y$

time taken to meet = $\frac{d}{x+y}$.

(ii) Same direction:-

Relative Speed in same direction.

R.S = $x-y$ [opposite direction]

time taken to cross/overtake = $\frac{d}{x-y}$.

$$t_m = \frac{d}{x+y} \quad \& \quad t_{o/c} = \frac{d}{x-y}$$